Systems of Linear Equations

Three elementary row operations:

① Swap the rows: $R_i \leftrightarrow R_j$.

② Scale a row by nonzero scalar $\lambda \in F$: $R_i \to \lambda R_i$.

③ Add multiple of row j to row i: $R_i \to R_i + \lambda R_j$

<u>Example.</u> Solve the following over $\mathbb{Z}_7$.

$$\left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -1 & 1 & 0 & -1 & 0 \\ 1 & 5 & 1 & 3 & 3 & 3 \end{array}\right) = \left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 6 & 1 & 0 & 6 & 0 \\ 1 & 5 & 1 & 3 & 3 & 3 \end{array}\right)$$

$$\xrightarrow[R_3 \mapsto R_3 - R_1]{R_2 \mapsto R_2 - R_1} \left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 2 \\ 0 & 5 & 0 & 6 & 5 & 5 \\ 0 & 4 & 0 & 2 & 2 & 1 \end{array}\right)$$

$$\xrightarrow{R_3 \mapsto R_3 + 2R_2} \left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 2 \\ 0 & 5 & 0 & 6 & 5 & 5 \\ 0 & 0 & 0 & 0 & 5 & 4 \end{array}\right)$$

$$\xrightarrow[R_3 \mapsto 3R_3]{R_2 \mapsto 3R_2} \left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 4 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 5 \end{array}\right)$$

$$\xrightarrow{R_1 \mapsto R_1 - R_2} \left(\begin{array}{ccccc|c} 1 & 0 & 1 & 4 & 0 & 1 \\ 0 & 1 & 0 & 4 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 5 \end{array}\right)$$

$$\xrightarrow{R_2 \mapsto R_2 - R_3} \left(\begin{array}{ccccc|c} 1 & 0 & 1 & 4 & 0 & 1 \\ 0 & 1 & 0 & 4 & 0 & 3 \\ 0 & 0 & 0 & 0 & 1 & 5 \end{array}\right)$$

$$\Rightarrow \begin{cases} x_1 + x_3 + 4x_4 = 1 \\ \quad x_2 + 4x_4 = 3 \\ \qquad\qquad x_5 = 5 \end{cases}$$

We call x_1, x_2, x_5 the leading or basic variables.
We call x_3, x_4 the non-leading or free variables.

We write our solution in terms of the free variables:

$x_1 = 1 - x_3 - 4x_4 = 1 + 6x_3 + 3x_4$
$x_2 = 3 - 4x_4 = 3 + 3x_4$
x_3 free
x_4 free
$x_5 = 5$

We can write the solution as

$$(x_1, x_2, x_3, x_4, x_5) = \{(1+6\lambda+3\mu,\ 3+3\mu,\ \lambda,\ \mu,\ 5) \mid \lambda, \mu \in \mathbb{Z}_7\}$$

or in vector notation,

$$\vec{x} = \begin{pmatrix} 1 \\ 3 \\ 0 \\ 0 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 3 \\ 3 \\ 0 \\ 1 \\ 0 \end{pmatrix}.$$

For each choice of $\lambda, \mu \in \mathbb{Z}_7$, we get a solution, thus, we have $7 \times 7 = 49$ solutions in total.

Behaviour of Solutions.

A system of linear equations can have

- <u>No solutions</u>: We call such a system inconsistent. This occurs if our augmented matrix has a row like
  $0 \cdots 0 \mid c$, where $c \neq 0$.

- <u>One solution</u>: occurs when we have no free variables, i.e. every column has a pivot.

- <u>Multiple solutions</u>: occurs when we have free variables.
  $\rightsquigarrow$ · F is finite: finite many solutions.
  · F is infinite: infinite many solutions.

Inverse and Elementary Matrices

Recall that a square matrix $A \in Mat_n(F)$ is invertible if $\exists B \in Mat_n(F) : AB = BA = I_n$. We write $B = A^{-1}$.

<u>Example.</u> $A = \begin{pmatrix} 2 & 4 & 0 \\ 4 & 2 & 4 \\ 0 & 2 & 4 \end{pmatrix} \in Mat_3(\mathbb{Z}_5)$

$$\left(\begin{array}{ccc|ccc} 2 & 4 & 0 & 1 & 0 & 0 \\ 4 & 2 & 4 & 0 & 1 & 0 \\ 0 & 2 & 4 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_3 \mapsto 3R_3]{R_1 \mapsto 3R_1} \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 3 & 0 & 0 \\ 4 & 2 & 4 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 3 \end{array}\right)$$

$$\xrightarrow{R_2 \mapsto R_2 + R_1} \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 3 & 0 & 0 \\ 0 & 4 & 4 & 3 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 3 \end{array}\right)$$

$$\xrightarrow{R_2 \mapsto R_2 + R_1} \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 & 3 \\ 0 & 1 & 2 & 0 & 0 & 3 \end{array}\right)$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 3 & 0 & 0 \\ 0 & 1 & 2 & 0 & 0 & 3 \\ 0 & 0 & 1 & 3 & 1 & 3 \end{array}\right)$$

$$\xrightarrow{R_2 \mapsto R_2 - 2R_3} \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 3 & 0 & 0 \\ 0 & 1 & 0 & 4 & 3 & 2 \\ 0 & 0 & 1 & 3 & 1 & 3 \end{array}\right)$$

$$\xrightarrow{R_1 \mapsto R_1 - 2R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 4 & 1 \\ 0 & 1 & 0 & 4 & 3 & 2 \\ 0 & 0 & 1 & 3 & 1 & 3 \end{array}\right) \Rightarrow A^{-1} = \begin{pmatrix} 0 & 4 & 1 \\ 4 & 3 & 2 \\ 3 & 1 & 3 \end{pmatrix}$$

Elementary Matrices

① For $i \neq j$, consider the matrix:

$$\begin{array}{cc} \begin{pmatrix} 1 & & & & & & & & \\ & \ddots & & & & & & & \\ & & 1 & & & & & & \\ & & & 0 & & & & 1 & \\ & & & & 1 & & & & \\ & & & & & \ddots & & & \\ & & & & & & 1 & & \\ & & & 1 & & & & 0 & \\ & & & & & & & & 1 \\ & & & & & & & & & \ddots \\ & & & & & & & & & & 1 \end{pmatrix} & \begin{array}{c} \\ \\ \\ i \\ \\ \\ \\ j \\ \\ \\ \\ \end{array} \\ \begin{array}{cccccccccc} & & & i & & & & j & & \end{array} & \end{array}$$

Multiply on the left by this matrix performs $R_i \leftrightarrow R_j$.

② For any i, $\lambda \neq 0$, consider the matrix:

$$\begin{array}{cc} \begin{pmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & & \\ & & & \lambda & & \\ & & & & 1 & \\ & & & & & \ddots \\ & & & & & & 1 \end{pmatrix} & \begin{array}{c} \\ \\ \\ i \\ \\ \\ \\ \end{array} \\ \begin{array}{ccccccc} & & & i & & & \end{array} & \end{array}$$

Multiplying on the left by this matrix performs $R_i \mapsto \lambda R_i$

③ For $i \neq j$ consider the matrix

$i < j$

$$\begin{array}{cc} \begin{pmatrix} 1 & & & & & & & & \\ & \ddots & & & & & & & \\ & & 1 & & & & & & \\ & & & \lambda & & & & & \\ & & & & \ddots & & & & \\ & & & & & 1 & & & \\ & & & & & & \ddots & & \\ & & & & & & & 1 & \\ & & & & & & & & \ddots \\ & & & & & & & & & 1 \end{pmatrix} & \begin{array}{c} \\ \\ \\ i \\ \\ \\ \\ j \\ \\ \\ \end{array} \\ \begin{array}{cccccccccc} & & & i & & & & j & & \end{array} & \end{array}$$

$j < i$

$$\begin{array}{cc} \begin{pmatrix} 1 & & & & & & & & \\ & \ddots & & & & & & & \\ & & 1 & & & & & & \\ & & & \ddots & & & & & \\ & & & & 1 & & & & \\ & & & & & \lambda & & & \\ & & & & & & 1 & & \\ & & & & & & & \ddots & \\ & & & & & & & & 1 \end{pmatrix} & \begin{array}{c} \\ \\ \\ j \\ \\ \\ \\ i \\ \\ \\ \end{array} \\ \begin{array}{cccccccccc} & & & j & & & & i & & \end{array} & \end{array}$$

Multiplying on the left by this matrix performs $R_i \mapsto R_i + \lambda R_j$

<u>Example.</u> $\begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}\begin{pmatrix} 3 & 2 \\ -1 & 4 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 8 & 10 \end{pmatrix}$

$R_2 \mapsto R_2 + 3R_1$

<u>Invertibility</u>

The row operations are invertible.

$$R_i \leftrightarrow R_j \rightsquigarrow R_i \leftrightarrow R_j$$
$$R_i \mapsto \lambda R_i \rightsquigarrow R_i \mapsto \tfrac{1}{\lambda} R_i$$
$$R_i \mapsto R_i + \lambda R_j \rightsquigarrow R_i \mapsto R_i - \lambda R_j .$$

Thus, the elementary matrices are invertible.

<u>Theorem.</u> A matrix A is invertible iff it can be row reduced to the identity.

Proof. Suppose A can be row reduced to the identity, i.e. $\exists\, E_1, E_2, \ldots, E_k$ such that

$$(E_k \cdots E_2 E_1) A = I_n .$$

$\Rightarrow$ By definition, A is invertible, and

$$\boxed{A^{-1} = E_k \cdots E_2 E_1 .}$$

Now suppose that A is invertible and let M be the unique matrix in RREF with $A \sim M$. Then $M = E_k E_{k-1} \cdots E_1 A$.

Since A is invertible and each E_i is also invertible, M must be invertible. Since M is invertible, it cannot have a row of all zeros. But the only matrix in RREF with no zero rows is $M = I$.

$\Rightarrow A \sim I$.